Guidance on CoBit: classifier-free, auto- and self-guidance

TinyGSM → GSM8K, production 500k bits model. Every guidance experiment we ran: what worked, what did not, why, and what to keep. Written 2 Oct 2026 from results on disk; no GPU was used.

> **Bottom line.** With deterministic sampling, CFG, AutoGuidance and SG-prev all help and SG-exact hurts. **CFG is best** (13.8% → 22.0%, +8.1 pp, at 2× compute), AutoGuidance gives +6.6 pp (2× compute plus a second model) and SG-prev +4.8 pp at no extra compute. But a **free sampler setting, stochastic churn, gives +12 pp on its own** (13.7% → 25.7%) and beats every guided deterministic run. Under churn, CFG only helps if re-tuned down to w≈2 (+1.9 pp), AutoGuidance turns harmful and SG-prev collapses. In the strongest regime nothing helps. Combining methods always hurts. Spending the extra compute on a second sample and a vote is as good or better. **The study is closed**; keep churn, voting, and CFG w≈2 only when a single sample is required.

1. The three methods in one page

All three take the model's prediction of the clean bits, D, and push it away from a "worse" prediction: $D_{guided} = D_{worse} + w \cdot (D_{good} - D_{worse})$. They differ only in what "worse" means. Because the score is linear in D, guiding D is the same as guiding the score; the analytic matched-filter term cancels out of the difference, so only the learned part is amplified.

Method	The "worse" prediction	Extra cost	Notes
CFG (classifier-free)	the same model with the question hidden (unconditional; trained by dropping the prompt 10% of the time)	2× network evaluations	w=1 is no guidance; w<1 moves *towards* the unconditional model
AG (AutoGuidance)	an earlier, weaker checkpoint of the same run (250k, 350k or 425k steps vs the 500k model)	2× evaluations + a second 2 GB model	only 250k–425k checkpoints were available
SG-exact (self-guidance)	the same model evaluated at a noisier level, $\sigma \cdot e^{0.5}$	2× evaluations	the matched filter is held fixed ("hold" mode)
SG-prev	the prediction cached from the previous sampling step	none	effectively a step-to-step extrapolation (momentum) term, ~33–96× amplified

Setup. One model: the production conditional run (`cobit_raw_binary_bits_cfg`, 500k steps, EMA), the only CoBit model trained with prompt dropout. The model writes a Python function; we run it and compare the returned number with the gold answer. Screens: the first 250 problems, 1 seed, 256 steps. Confirmations: all 1319 problems × 3 seeds, 512 steps, paired bootstrap over problems. All on CSD3 A100s. Three sampler regimes:

- **Regime A, deterministic** (γ=0): DDIM on the entropy-rate schedule. Baseline 13.8% (512 steps).
- **Regime A with churn** (γ=0.2–0.3): fresh noise is injected at each step (EDM "churn"). Same cost, much higher baseline.
- **Regime B** (γ=0.41, 1024 steps): the collaborator's strongest setting, baseline 29.6%.

Note: the often-quoted 0.164 baseline is the 250-problem screen only; those problems are about 4 points easier than the full set. All conclusions below use full-set numbers unless marked n=250.

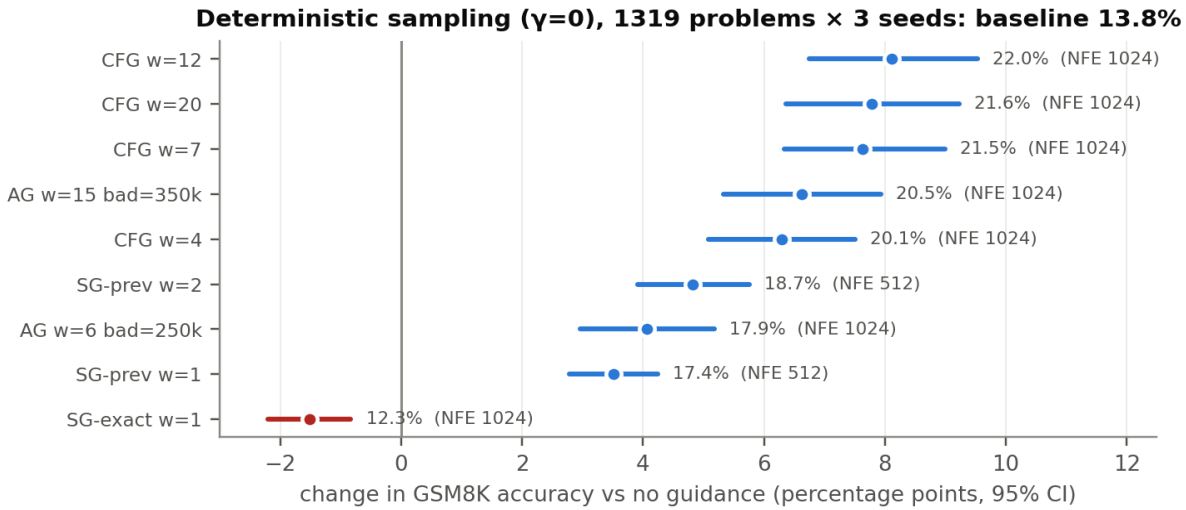
2. Deterministic sampling: guidance works

Screens (n=250): finding each method's operating point

CFG scale w	0.25	0.5	0.75	1	1.25	1.5	2	3	4
accuracy	0.0%	0.0%	12.8%	16.4%	14.0%	16.8%	18.8%	19.6%	21.6%
CFG scale w	5	7	8	10	12	15	20	30	
accuracy	21.2%	22.4%	22.0%	22.4%	22.8%	24.4%	25.2%	24.4%	

- **CFG** rises from w=1 and plateaus above w≈7, with no turnover up to w=30. Below w=1 it is catastrophic (0% at w=0.25–0.5), because it moves towards an unconditional model that cannot do the problem.
- **AG**: the 350k "bad" model is best and 425k worst at every scale. A bad model too similar to the good one gives a tiny, noisy difference. Only 2 of ~30 screen cells were individually significant, roughly chance.
- **SG-prev** helps at w=1–2 at every step count; **SG-exact** gets worse as its scale grows, at every step count. Holding the matched filter fixed beats varying it, more so at larger scale.
- The **null prompt** used for the unconditional branch barely matters (22.8% / 22.8% / 21.6%).
- **Steps:** the baseline plateaus near 17% (n=250) whatever the step count; CFG at only 8 steps already beats the baseline at any step count. Guidance buys a higher plateau, not a faster one.

Confirmation on the full test set



Source: results/guidance/confirm_summary.json (n=1319, 3 seeds, 512 steps). All p < 1e-4. Re-scoring on the 1069 problems never used for selection moves every number by ≤ 0.2 pp.

The CFG optimum is a plateau (w=7, 12 and 20 are within noise of each other). SG-prev's gain is not a better integrator in disguise: a second-order Heun solver at the same cost gains nothing (−0.1 pp), and SG-prev beats it by +4.9 pp [+4.0, +5.9].

Is it worth the compute?

CFG and AG double the cost. The fair comparison spends that budget on a second sample instead (numbers from GUIDANCE_REPORT.md and commit cdde4c6; voting keeps only programs that run):

At γ=0, 1024 network evaluations each	Accuracy / change
Best of 2 samples (oracle, pass@2)	20.1%
Majority vote of 2 samples (deployable)	18.0%
CFG w=12 vs oracle pass@2	+1.8 pp [+0.3, +3.4]
CFG w=12 vs vote of 2	+3.3 pp [+2.0, +4.5]
AG w=15 vs oracle pass@2	+0.4 pp [−1.1, +1.8] (not significant)
AG w=15 vs vote of 2	+1.8 pp [+0.6, +3.0]

So in the deterministic regime CFG is genuinely worth its compute; AG only beats the deployable vote.

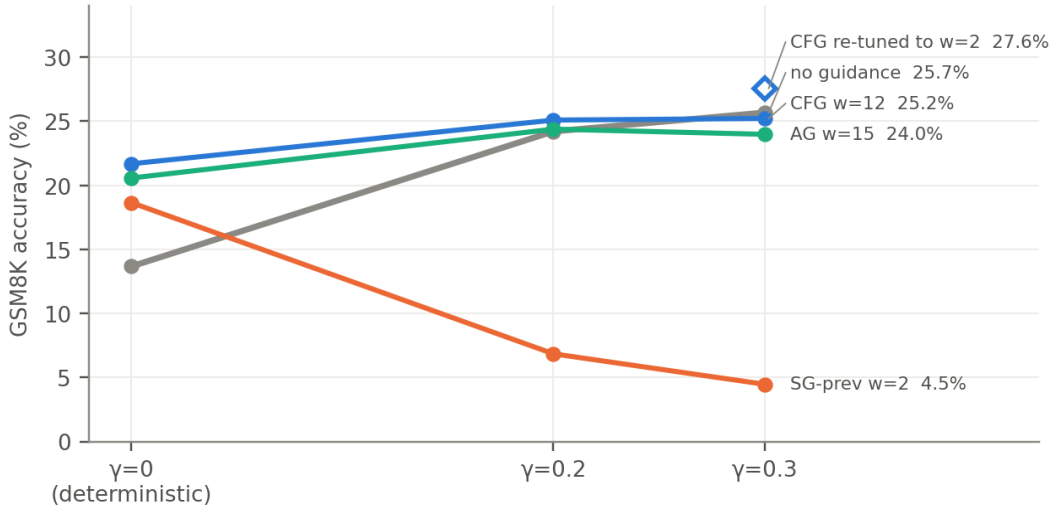
Combining methods: always worse

Cell (2×2×3 factorial, n=1319, 3 seeds)	Accuracy	NFE
CFG w=12	22.0%	1024
AG w=15	20.5%	1024
AG + SG-prev	20.0%	1024
SG-prev w=2	18.7%	512
no guidance	13.8%	512
CFG + SG-prev	12.6%	1024
SG-exact w=2	11.7%	1024
CFG + SG-exact	7.7%	2048
CFG + AG	0.1%	2048
CFG + AG + SG-prev	0.0%	2048

Every combination is worse than its best component, and CFG+AG collapses to 0%: stacking two amplifications drives the bits to hard 0/1 and the programs stop being code.

3. With churn: the picture reverses

Adding sampler noise (churn) helps more than any guidance, and reverses most of it



n=1319, 3 seeds, 256 steps (so γ=0 here is 13.7%, not the 13.8% at 512 steps). Source: results/guidance/all_cells.csv, grids stoch_confirm and churn_anatomy.

Change vs the same-γ baseline (pp, 95% CI)	γ = 0	γ = 0.2	γ = 0.3
CFG w=12	+8.0 [+6.7, +9.4]	+0.9 [−0.6, +2.3]	−0.5 [−1.8, +0.9]
CFG re-tuned w=2 / w=4	—	—	**+1.9 [+0.7, +3.2] / +1.7 [+0.5, +3.0]**

Change vs the same-γ baseline (pp, 95% CI)	γ = 0	γ = 0.2	γ = 0.3
AG w=15	+6.9 [+5.6, +8.2]	+0.2 [−1.2, +1.6]	−1.7 [−3.1, −0.3]
SG-prev w=2	+5.0 [+4.1, +5.9]	−17.4 [−19.0, −15.8]	−21.2 [−23.1, −19.5]
SG-prev, smallest tried (w=0.06)	—	—	−1.3 [−2.5, −0.2]
Churn itself (no guidance, vs γ=0)	—	**+10.5 [+9.2, +12.0]**	**+12.0 [+10.6, +13.5]**

CIs from docs/findings.md (F4–F7) and docs/report/numbers.json.

- **Churn is the single largest effect in the study**, at zero extra cost: the unguided γ=0.3 baseline beats CFG w=12 at γ=0 by +4.0 pp [+2.6, +5.4].
- **CFG was mis-tuned, not made useless:** its best scale moves from 12 down to 2–4 and its gain shrinks from +8 to +2 pp.
- **AG** becomes harmful (only w=15 was tested under churn). **SG-prev** fails at every scale, and the damage grows linearly with w, which points to a biased direction rather than noise.
- **Equal compute at γ=0.3:** a 2-sample vote without guidance (28.8%, 512 NFE) is at least as good as CFG w=2 (27.6%, 512 NFE): −1.2 pp [−2.7, +0.2] for CFG. Voting adds about +3 pp on top of either, so the two stack.
- **Regime B** (γ=0.41, 1024 steps; screen of n=250, 1 seed): nothing helps. Best is CFG w=2 at +0.4 pp [−3.6, +4.4]; AG w=15 −9.2, SG-exact −14.8, SG-prev w=2 goes to 0%.

4. Why: the mechanisms we found

Guidance fixes some answers and breaks others, at fixed rates

Method (γ=0, n=1319, 3 seeds)	Fixes (of wrong answers)	Breaks (of right answers)	Break-even accuracy a*
CFG w=12	13.5%	25.5%	**35%** [31, 39]
CFG w=7	12.6%	23.4%	**35%** [31, 39]
CFG w=4	10.7%	21.4%	**33%** [30, 38]
SG-prev w=2	7.6%	12.2%	**38%** [33, 45]
AG w=15	12.2%	28.1%	**30%** [27, 34]
AG w=6 (250k bad model)	9.2%	28.1%	**25%** [22, 28]

Guidance is not a targeted correction: CFG rescues ~10–14% of wrong answers and destroys ~20–25% of right ones, in every regime we checked. The net effect is arithmetic: gain = fix·(1 − a) − break·a, which is positive only while the baseline accuracy a is **below** a* = fix/(fix + break) ≈ 33–35%. At 13.8% there are many wrong answers to fix, so guidance wins; churn raises the baseline to 25–30%, close to a*, and the margin disappears. (The 9 Sep meeting report states this the wrong way round, "only positive above ≈33%"; the correct direction is *below*.)

Guidance damages code validity, not arithmetic

In the Regime B error analysis (n=250), the number of programs that do not run rises with guidance strength (98 baseline → 102 CFG w=2 → 126 AG w=15 → 166 SG-exact → 250 SG-prev w=2), while "runs but wrong" falls. Guidance mostly breaks answers by making the code invalid.

Why each method fails where it does

- **SG-prev under churn:** the cached prediction comes from the noisier state *before* the churn kick, so the difference points back toward a noisier estimate. Its size grows ~16× with churn while the step spacing is unchanged. Not yet tested directly.

- **SG-prev at w=2 in Regime B:** a runaway trajectory, not a decoding failure. The state becomes anomalous by step ~88; guidance and score then inflate together (positive feedback), and the output becomes token soup with no `def` in 100/100 samples.
- **AG:** needs a bad model that is different enough (350k) but not useless; under churn its correction adds harm.
- **SG-exact:** harmful in every regime; the one robust negative.
- **Overall pattern:** guidance's value is concentrated where the sampler is weak. Once the sampler is improved (churn), most of guidance's benefit is already captured.

5. What to keep

> **Keep:** churn (γ≈0.3) as the default sampler setting; **majority voting over programs that run** (the best known configuration is no guidance, γ=0.3, vote of 2: 28.8% at 512 NFE); CFG at w≈2 with churn only when a single sample is required (27.6%); the "hold" matched-filter mode; and the habit of quoting every scale together with its sampler regime.
> **Drop:** SG-exact (any scale), SG-prev (anywhere churn is used), AutoGuidance (needs a second model and turns harmful with churn), CFG w<1, and every combination of methods.

6. Corrections to earlier write-ups

- **0.164 is not the full-set baseline.** It is the first 250 problems, which are easier. The full-set deterministic baseline is 13.7% (256 steps) / 13.8% (512 steps).
- **Sign of the break-even rule** in docs/report_meeting/report.tex:87: positive *below* ≈33%, not above (correction drafted in docs/cpu_reanalysis.md, awaiting approval).
- **GUIDANCE_REPORT.md §6–9 is superseded** by docs/findings.md (F2, F5–F8): the factorial, solver and compute controls did run, and its recommendations do not hold once churn is on.
- **23.4% vs ~29%** are different models and samplers (base run at 250k with churn γ=0.34, vs 425k with FKC churn 0.41), not a discrepancy. The ~29% replicates (28.1% on all 1319 problems); the gap to our 13.8% is churn, not steps or precision.
- The factorial "interaction −0.166" in docs/report/report.tex does not match any value on disk; docs/cross_regime_comparison.md mixes 256- and 512-step numbers in its γ=0 column.

7. Open questions (not pursued)

- Why does churn help so much? It is the most valuable open question and is not about guidance.
- Test the SG-prev-under-churn explanation by recomputing the cached prediction after the churn kick.
- Larger votes under churn (maj@4 vs CFG w=2 maj@2 at equal compute), ~0.15 GPU-h.
- Caveats: one task, one model, one sampler family; differences below ~2 pp are not resolvable at n=1319; Regime B is n=250 with one seed; the production model itself carries the multi-GPU σ_{data} training bug (T1), whose training-time effect is untested.

Sources. docs/GUIDANCE.md, GUIDANCE_REPORT.md, findings.md, open_questions.md, cross_regime_comparison.md, tiny_gsm8k_error_analysis.md, trajectory_dynamics.md, replication_audit.md, best_baselines.md, report/report.tex, report_meeting/report.tex, cpu_reanalysis.md; diffusion/continuous/guidance.py; results/guidance/all_cells.csv, confirm_summary.json, confirm_summary_holdout250.json, paired_comparisons.json; results/cpu_reanalysis/t6_fix_break.json, t4_voting.json.